David Smith

Computer Science & Economics Student

262-749-2376 smithdave2004@gmail.com linkedin.com/in/davidchrsmith

davidchrsmith.github.io/PortfolioWebsite

Education

University of Minnesota Twin-Cities **Expected May 2027**

Dual Degrees: B.S. in Computer Science and B.S. in Economics (GPA: 3.544) *Minneapolis, MN*

- **Relevant Coursework:** Advanced Algorithms, Software Engineering, Machine Learning, Database Systems, Secure Software Systems, Industrial Organization, Environmental Economics
- CSE Dean's List, Spring 2024, Spring 2025, and Spring 2026

Experience

HaFSBX **May 2026 – Present**

R&D Intern *Maplewood, MN*

- Engineered full-stack web applications using **React**, **Next.js**, **Django REST Framework**, **PostgreSQL**, and **Three.js** to deliver new features and improve application usability.
- Designed, implemented, and deployed ROS2 robotic applications using **Python**, **C++**, and **XML** launch/configuration files.
- Designed, implemented, and validated REST APIs and backend data pipelines using **OpenAPI** and **Bruno** for API design, documentation, and validation.
- Developed and validated robotic workflows in **NVIDIA Isaac Sim** before deployment through **Mission Control** and **Mission Dispatch**.

Lavner Education **June 2025 – Aug. 2025**

Internship - STEM Instructor *St. Paul, MN*

- Taught and mentored students (ages 6–14) in Python, C++, robotics, and AI Ethics, fostering problem-solving and teamwork skills.
- Collaborated with instructors to refine curriculum, adding interactive coding challenges and hands-on projects that improved student engagement and project completion rates.
- Debugged student code and robotics programs in real time, providing individualized guidance and reinforcing core programming concepts.
- Troubleshot technical issues with laptops, IDEs, and robotics kits, minimizing class disruptions and ensuring smooth delivery of lessons.

Projects

Personal Portfolio Website

- Built a responsive portfolio website using **React**, **JavaScript**, and **CSS** to showcase technical projects through an interactive, modern interface.
- Implemented interactive 3D graphics and smooth animations using **Three.js**, creating an engaging user experience.

Drone Package Delivery Simulation

- Implemented a drone delivery simulator in **C++** with **Docker** containerization, simulated package theft scenarios, and a notification system for completed deliveries.
- Developed multiple path-finding algorithms for route optimization and evaluated efficiency across strategies.
- Applied object-oriented design principles to model autonomous drone behavior and real-world delivery scenarios.

Invasive Species Management Cost-Benefit Analysis

- Conducted a comprehensive cost-benefit analysis of Early Detection and Rapid Response (EDRR) strategies for invasive species management in the Twin Cities, evaluating municipal costs, long-term economic benefits, and policy tradeoffs using **NPV**, **IRR**, and sensitivity analysis.
- Developed data-driven policy recommendations and presented findings to the Twin Cities Metropolitan Council, demonstrating the financial viability of proactive invasive species management for municipal decision-making.

Technical Skills

Languages: Python, C++, C, Java, JavaScript, SQL, OCaml
Frameworks: React, Next.js, React Native, Django, Django REST Framework, Three.js
Robotics: ROS2, NVIDIA Isaac Sim, Foxglove
Tools: Git, GitHub, Docker, Linux, PostgreSQL, Bruno, Figma, RStudio, OpenAPI